Ref. Ares(2019)6451597 - 18/10/2019

ESSnet Big Data II

Grant Agreement Number: (847375-2018-NL-BIGDATA)

https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata
https://ec.europa.eu/eurostat/cros/content/essnetbigdata_en

Workpackage J

Innovative Tourism Statistics

Deliverable J1: ESSnet Methods for web scraping, data processing and analyses.

Final version, 23.07.2019

Prepared by:

WPJ Team

Workpackage Leader:

Marek Cierpiał-Wolan (Statistics Poland, Poland)
M.Cierpial-Wolan@stat.gov.pl
telephone : + 48 17 85 35 210 ext. 311
mobile phone : + 48 515 037 640

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Introduction

The general objective of the ESSnet Big Data I was to prepare the European Statistical System (ESS) for implementation of big data sources into the production of official statistics. Due to the diversity of those sources it explored aspects of data access and use, development of methodology, quality and technology, processing and dissemination and production of experimental outputs. The research result allowed this step to be taken for some of the big data sources and statistical domains. However, several other potentially relevant big data sources have not been investigated for the ESS.

The project ESSnet Big Data II aims at continuing the preparation of the ESS for integration of big data sources into the production of official statistics. Considering the nature and diversity of big data sources, this objective requires exploring the possibilities of using these sources as well as dealing with arising difficulties. Following the guidelines of the European Commission, ESSnet Big Data II includes new pilot projects that should result in experimental statistics[1] and in consequence be implemented in the statistical production.

One of the pilot projects is the workpackage J - "Innovative tourism statistics" (WPJ), whose principal purpose is to address the need of a conceptual framework and setting up a smart pilot Tourism Information System that will support statistical production in the field of tourism by integrating various big data sources with administrative registers and statistical databases using innovative statistical methods.

Information obtained through web scraping will be a key element in building a prototype of Tourism Information System, which will allow us to combine big data sources, administrative data and official statistical data. This system can be used in official statistical production to improve the quality and completeness of data.

The main objective of the report is to describe and analyse web scraping methods as well as to evaluate the extracted data.

The specific objectives of the document are:

1. Presentation of a new data acquisition model using web scraping tools.
2. Initial assessment of the effectiveness of the implementation of new solutions on selected portals.
3. Evaluation of the quality of variables using statistical tools.

This document consists of the following chapters:

Chapter 1 presents a short review of the literature related to mainly web scraping model that can be used to obtain data from web portals.

Chapter 2 is devoted to the methods and the next stages of the web scraping process. In particular:

[1] Experimental statistics are compiled from new data sources and methods in an effort to respond to users' needs. It should be noted that these statistics have not reached full maturity in terms of harmonisation, coverage or methodology. [https://ec.europa.eu/eurostat/web/experimental-statistics]

- subchapter 2.1. describes the concept adopted by the WPJ for obtaining data from online portals containing information in the field of tourism, along with the actions to be taken at each of the specified stages;
- subchapter 2.2. includes discussion on the preparation of a catalogue of identified websites and their analysis in terms of the use of various web scraping technologies;
- subchapter 2.3. discusses the aspects of the legality of the web scraping process;
- subchapters 2.4. and 2.5. present general and detailed approaches to obtaining data using web scraping together with a description of the new solution proposed by WPJ for the needs of the project.

Chapters 3 and 4 relate to the organization of the processing (cleaning and standardization of data) and the analysis of data obtained from selected websites as part of the first phase of the implemented work. They are also the introduction to activities that will be continued further in the next periods of the project. Their preliminary results will be presented in the next report "J2- Interim technical report showing the preliminary results and a general description of the methods used".

The work was carried out under the WPJ for the first five months of the task implementation (November 2018 to March 2019).

1. Literature overview

An overview of the literature within the ESSnet Big Data II project is prepared by the workpackage K, whose main task is to consolidate knowledge gained in this ESSnet in the area of methodology and quality when using big data in the statistical production process. Therefore, this chapter focuses only on the overview of the literature related to the work of the WPJ conducted in the field of web scraping, in particular concerning the development of a universal data extraction model.

Data sources used to generate results in the field of tourism statistics do not always give a full picture of the phenomenon. In recent years, many websites have been created containing information that could benefit official statistics in this area for estimating the number of tourists, travel costs, as well as in determining the most popular tourist destinations, among other things.

One of the methods used to obtain data from tourist websites is the web scraping method. The effectiveness of this method depends mainly on the systematization of the whole process. Work on creating a data extraction model with the use of web scraping tools was carried out, among others, by Abburu and Babu [2013]. The authors created a model consisting of three steps: searching for URLs of websites, extracting information from websites and data analysis. Another model of data extraction is described by Schmunk et al. [2014], proposing to divide it into five stages. According to the authors, the first step is to select and catalogue portals according to the relevant areas and then to conduct the structural analysis of selected portals. The next step is the process of data extraction, followed by their evaluation and use in official statistics.

Based on the analysis of the results obtained by the above-mentioned authors and own experience, it was found that the data extraction model should be more comprehensive.

One of the key stages of the model is the web scraping process. The dynamic development of software used to create and modify websites forces programmers to create tools for downloading data, improving the web scraping methods.

The variety of techniques and methods used when building websites is a problem for the creation of universal methods of web scraping. The basic two ways to access the content of websites that users use are: access through Application Programming Interface (API) or HyperText Markup Language (HTML) extraction. In the first way, website administrators provide a set of methods to users for promotional purposes, although often this set is limited in time [Glez-Pena et al. 2014]. Increasing the limits of API use depends on the financial possibilities of users interested in extracting data. The main advantage of this method is obtaining information that is already cleared of unnecessary elements and normalized, e.g. Extensible Markup Language (XML) or JavaScript Object Notation (JSON). The HTML extraction method involves scraping of the content available on websites. When scraping, the text is automatically processed using both the coded of logic interaction with the service and semantic interpretation.

There are several web scraping techniques. The simplest ones are copy/paste as well as text pattern matching. The copy/paste technique consists in manually searching for information and pasting it into a catalogue or document, while text pattern matching uses regular expressions for HTML text processing.

Other techniques, e.g. Hypertext Transfer Protocol (HTTP) programming and Document Object Model (DOM) parsing, require more knowledge and experience. Using the HTTP programming technique, dynamic and static pages can be processed by sending HTTP requests to remote servers. This method uses socket programming, which allows to extract more data in less time although it is more complicated in terms of implementation [Johansson et al. 2015]. The DOM parsing technique involves the extraction of data using the built-in software in the web browser. This software will break down the structural elements of the HTML language of the website or the DOM model. The method is used in many research projects because of its effectiveness [Cai et al. 2003].

Currently, the most advanced approach is simulated user interaction with the service using screen-scraping. It allows full interaction with selected websites, based on dynamically modifying the DOM tree, Cascading Style Sheets (CSS) components and JavaScript [Umamageswari and Kalpana 2017]. The most important advantages of this solution are imitating the website user behaviour, the ability to build interaction scenarios, storing archived copies of data for later processing as well as the possibility to act concurrently.

2. Methods

2.1 Web scraping – flow model

In order to collect data from various Internet sources, WPJ created its own concept of a web scraping process for the field of tourism. It consists of three stages:

Stage 1: Methods for web scraping to gather the INPUT data, i.e.

- identification of websites,
- analysis of the catalogue of websites,
- analysis of legal aspects.

Note that the collected data are not structured and not suitable for immediate analysis.

Stage 2: Data processing steps to PROCESS data, i.e.

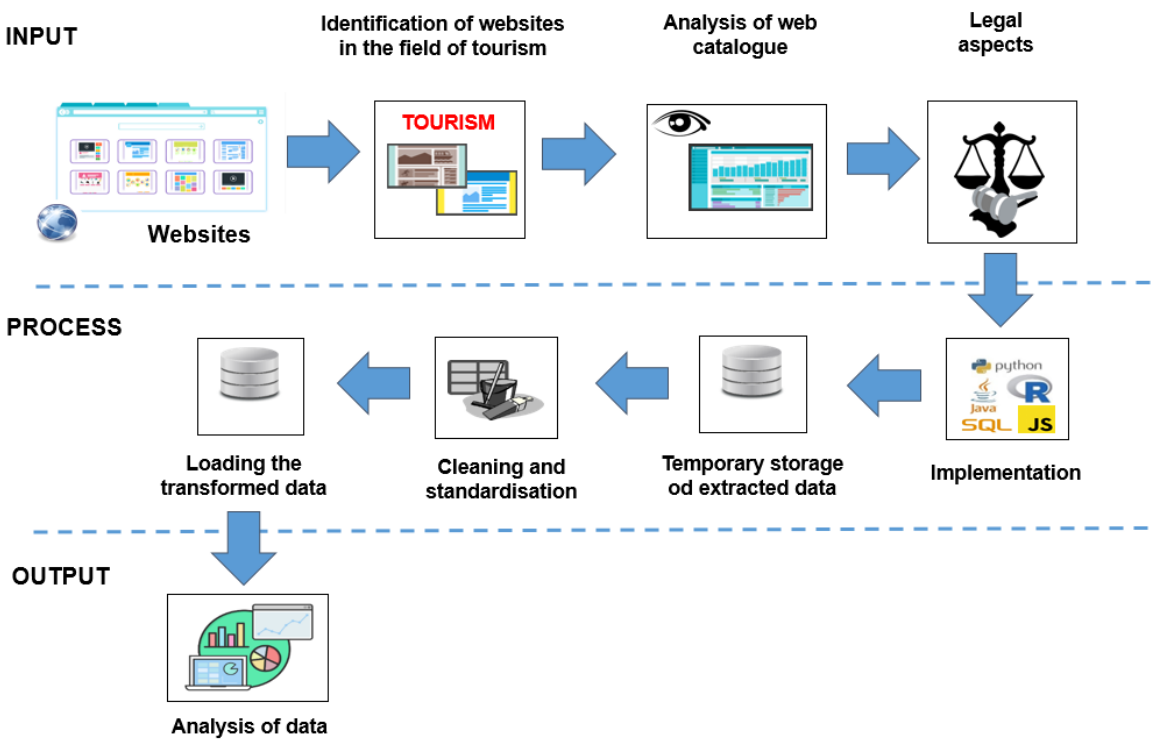
- implementation and saving of extracted data in a temporary database,
- cleaning and standardization,
- loading the transformed data into the data warehouse.

Stage 3: Data analysis, final stage to generate OUTPUT data:

- analysis of the collected data.

The diagram of the proposed web scraping process is depicted in Figure 1.

Figure 1. Diagram of the web scraping process



Source: Own work.

2.2 Identification and analysis of the Internet portals catalogue

The first step in the proposed method of data extraction is the identification of websites containing information on transport, accommodation and other services related to tourism followed by the separation of the catalogue of portals offering these services. Then it is necessary to analyse the created catalogue of websites in terms of the possibility of using web scraping methods.

Simple static pages do not require advanced tools to download such data. Unfortunately, more developed websites have a more complex structure and thus also require a more complex service. For simple pages, the process of data acquisition is easy and can be implemented in a few steps:

- preliminary analysis of the page markup structure,
- downloading the structure and data of the page in the form of a DOM tree i.e. hierarchically ordered elements of websites,
- identifying tags and HTML styles that contain useful data,
- extraction of data from HTML to text,
- saving the data in the target format, e.g. text file (TXT), comma-separated value format (CSV).

In the case of more complex pages that often use different technologies the task is more complicated and only general steps can be provided:

- initial analysis of page markup structure, analysis of used programming libraries, type of website (one-page, multiple-page, static, dynamic) application of synchronous/asynchronous technologies (a synchronous request blocks the client until operation is completed while an asynchronous request doesn't block the client and user can perform another operations), etc.,
- identification of tags on pages containing useful data,
- analysis of data availability in offline mode, necessity of user interaction,
- analysis of data availability in the DOM tree (waiting for all data to be loaded, exceptions handling),
- support for navigation mechanisms, paging,
- loading of subpages and processing and extraction of selected data,
- use of session variables/cookies,
- errors and exceptions handling related to the incorrect structure of HTML codes,
- conversion of data to a uniform character encoding standard.

An important element of the analysis process is the verification aimed at assessing the portal in terms of coverage, i.e. the division of portals into two types: global and local. Thus, data collected by means of web scraping may belong to the information layer available in all European Union countries, i.e. on a global scale or to the local data layer on tourism within particular countries. Regardless of the affiliation to a layer, selected portals should be characterized by:

- a large amount of indexed data,
- high detail of data and frequent updating,

- high activity and interaction of users,
- usability to enrich the results of traditional statistics.

WPJ is working to identify portals that can have an impact on tourism statistics. Examples of such portals are presented in Table 1.

Table 1. Classification of portals by type

Type	Name of the portal
global	www.hotels.com www.booking.com www.kayak.com www.tripadvisor.com www.airbnb.com www.zomato.com www.zoover.nl www.belvilla.nl
local	www.noclegi.pl www.hrs.de www.holidaycheck.de www.pincamp.de www.hotel.idealo.de www.hoevelogies.nl www.megaubytovanie.sk www.ubytujsa.sk www.pampeliska.cz www.bghotelite.com

Source: Own work.

During the project work the above mentioned portals were analysed in terms of the number of offers and variables available, and could be obtained in the process of web scraping to improve the production of statistics in the field of tourism (Table 2). Variables such as the type of accommodation and the number of beds were identified as relevant to the supply side of tourism, whereas the prices of accommodation, transport, meals and other services related to tourists' trips as the ones relevant to the demand side.

The data obtained in this way will be used to improve and enrich the existing system of tourism statistics in respect of both supply and demand side of tourism.

Table 2. Classification of portals by information possible to obtain

Information	Name of the portal
Accommodation (supply and demand)	www.hotels.com www.booking.com www.kayak.com www.tripadvisor.com www.airbnb.com www.zoover.nl
Accommodation (supply and demand)	www.belvilla.nl www.noclegi.pl www.hrs.de www.holidaycheck.de www.pincamp.de www.hotel.idealo.de www.hoevelogies.nl www.megaubytovanie.sk www.ubytujsa.sk www.pampeliska.cz www.bghotelite.com
Transport	www.booking.com www.kayak.com www.tripadvisor.com www.holidaycheck.de www.hotel.idealo.de
Food	www.tripadvisor.com www.zomato.com

Source: Own work.

At the initial stage of the work, hotels.com and booking.com were chosen from the above mentioned portals and scripts were prepared for them. In the next stage of the project, subsequent pages containing information on the accommodation base (including: booking.com, www.kayak.com, www.airbnb.com), as well as on the travel of tourists and their expenses on transport, food and other services (including www.expedia.com, www.tripadvisor.com, www.365tickets.com) will be subject to the web scraping process.

A comparative analysis is carried out systematically between the catalogue of accommodation establishments obtained from web scraping and the survey for tourism accommodation base in order to determine the representativeness of the pages.

2.3 Legality of web scraping and netiquette

Before starting the process of extracting data from web portals related to tourism, the legal aspects of web scraping should be taken into account [Stateva et al. 2017]. The legitimacy of using web scraping raises some doubts. In the regulations of many websites, provisions prohibiting this type of practice can often be found. All European Union countries have ratified the provisions regarding the protection of personal data and the protection of copyrights. In some countries great emphasis was also placed on the development of ethical codes.

During the work carried out in Big Data I the problem of the legality of extracting data by means of web scraping was addressed. Based on the analysis of the regulations, there are no legal restrictions concerning activities related to web scraping of websites. All data collected by this method are, however, subject to statistical confidentiality. The downloaded data should be analysed and stored in a safe environment and should be subject to strict control and verification before publication.

Another solution to the problems with legality of web scraping could be establishing permanent cooperation between official statistics and websites administrators in the area of information needs. In exchange for information data administrators could be presented with extracted and aggregated data valuable for the purpose of their activity.

However, this is a temporary solution because the constantly growing number of portals prevents a direct contact with the websites administrators due to the time constraints. Therefore, it is of great importance to create a common act of law that could regulate the above mentioned issues. Until these laws exist one of the guidelines that can be followed is the recommendations set out in the document *Legal aspects related to Web scraping of Enterprise Web Sites* developed as part of the Webscraping/Enterprise Characteristics package of Big Data I.

Therefore, during the web scraping process, as well as after gathering the necessary information, WPJ is following the obligation to maintain statistical confidentiality as well as the provisions regarding the protection of personal data and the protection of copyrights, among other things. Good web scraping practices are also preserved. Before starting the web scraping process of a chosen portal offering tourist services, the robots.txt file, which is provided by the website owner and contains information about portal elements that can/cannot be scraped, is checked. Particular attention is also paid to the traffic on the web scraped portals. In order to avoid overloading, appropriate delays are applied in server requests (http request). Data are also collected during off-peak hours, so that the owners are not exposed to unnecessary costs related to the acceleration of server work.

2.4 Recognition and evaluation of available solutions

A program for web scraping can be built in almost every programming language. Currently, the most popular languages or platforms are Python, R, Node.js, Ruby and Java. However, there is no unambiguous answer to the question: "which language is the best?". Regardless of the choice, a few features that help match the right language to the needs should be kept in mind, such as:

- flexibility,
- crawling efficiency,
- ease of coding,

- scalability,
- operational ability to feed database,
- active community.

The choice of tools often also depends on the structure of the portal and the type of data that need to be obtained. During the development of the web scraping process, the participants of WPJ extracted the most common methods of data acquisition, which are presented in Table 3.

Table 3. The most popular methods of data extraction

Web scraping method	Pros	Cons
Public API	The easiest way to extract data from the website. By providing a public API the data holder anticipates that others are interested in the data and may access it in an official way. API often comes with documentation. The data contained in JSON can be easily analysed.	The website owner decides what data can be extracted. A key used to identify the account is often required to send a request. Limiting the number of requests per period.
Hidden API	The method is resistant to HTML and CSS code changes. The data contained in JSON can be easily analysed.	It requires a lot of knowledge to check if the method is available. The website owner can disable the web services API, which can be used for web scraping. Bots can be blocked by monitoring excess traffic. Changes in the API are not documented.
HTML extraction	The method works very well on portals with a large amount of JavaScript. It gains in popularity and therefore has a very large and active community	Small changes in the HTML/CSS code surrounding important data and navigation elements would require more human involvement in the initial configuration or they may make the target website too difficult to web scrap. Bots can be blocked by monitoring excess traffic. Changes in the site's structure are not documented. High demand for CPU and RAM

Source: Own work.

2.5 New implementation solution

In the course of identifying and evaluating available web scraping solutions, the WPJ team proposed the introduction of an innovative method allowing for a more universal approach to web scraping and shortening the time spent on analysing the structure of portals.

Services that return data in a standard format such as CSV, XML and JSON are rare. At present, one-page services, i.e. dynamically reading content, are gaining great popularity. A significant number of websites use the latest solutions offered by Web 2.0 technology, such as dynamic content injection, modern Graphical User Interface (GUI) components, which require interaction from the user, protection against unauthorized access (Secure Socket Layer - SSL), page profiling for a certain group of browsers and testing their versions, use of graphical elements offered by HTML5 standard, e.g. for rendering charts, etc. Unfortunately, at present programming libraries that cover the whole spectrum of requirements for data processing by screen-scraping do not exist.

The new approach to the process of extracting data from web sources (real web pages or their copies in the form of HTML files) is based on the concept of simulated web browsing that imitates the user's behaviour.

This solution allows for a dynamic reading of page elements by automated simulation of the user’s behaviour occurring while browsing the portal. The application also ceases interactions with the portal until all page elements have been loaded, which reduces the portal analysis time needed to set appropriate delays and ensures that all information downloaded from the portal will be saved. Another very important advantage of the solution is minimizing the load on servers on which the web portal is located and checking the so-called "off-peak hours" because using the simulation of the user’s behaviour for servers, the tool behaves like just another person browsing.

It is worth emphasizing that also portals that have a static structure can be successfully processed with the use of the new approach, similarly as in the case of HTML extraction techniques, or through access to API.

The WPJ team developed a dedicated solution, which was distributed to all partners involved in this pilot project. At this stage the scripts go through the testing phase and corrections are made according to the exceptions reported by the partners. Moreover, testing of simulated interaction, parsing and extraction of data from the DOM tree is conducted. At the same time, reporting the indicated data to files for selected portals in order to verify the universality of the solution is being carried out. The results of these works, verification of the solution's potential and evaluation of the impact of the data obtained with this method on official statistics production in the field of tourism will be described in the final report.

3. Data processing

3.1 Cleaning

Web scrapped data will almost never be in a format suitable for immediate analysis. One of the most important stages in the web scraping process is therefore the processing and transformation of the extracted data in a specific way. This ensures a uniform format and links between the data, as well as the removal of redundant and duplicated information. Pre-treatment of the data is to remove any additional and unnecessary elements, such as unnecessary HTML code, accents, white characters, etc., which are not relevant. Often data extracted from different portals have different scope. It is possible to select information downloaded from portals in terms of completeness and suitability for the characteristics of specific variables or to use mechanisms allowing to supplement missing information with data downloaded from other portals.

3.2 Standardisation

The next step in the process is the standardisation of data. At his stage, the data obtained from different portals are transformed into a common format, which enables their comparison and analysis. During this process, character strings are converted to lowercase letters, dates and numeric values are formatted (gaps between fractions and separators of thousands), because each portal can save them in different formats. The conversion of currency in which amounts are collected from web scrapped pages is also important. Due to the fact that classifications identifying records differ from portal to portal (e.g. type of accommodation), it is necessary to prepare dictionaries for mapping and allowing conversion of the types used by portals to the types that will be used during the analysis. One of the most complicated operation in the standardisation process is formatting address data due to inconsistencies in naming, use of abbreviations and errors (typos) in addresses. Moreover, the same address can be written in many different ways. Identification and assignment of the actual address is impossible in many cases. The problem is also that the address data are not updated by the responsible entities, so some of the data collected may be outdated.

Further work focuses on defining the characteristics of a record that are important for determining its uniqueness. Selecting too simple signature will result in the loss of data, which were not necessarily duplicates, while defining a signature with too many features describing the record may leave in database records which are duplicates. Detection of duplicates is hampered by the fact that they are already working on standardised data, which also have a certain range of error.

During the work carried out under WPJ, data extracted by means of web scraping were stored in individual files referring to particular days. These data usually had one of the common formats, e.g. CSV or MS Excel format (XLSX). Collecting these data into a single database would have required a lot of effort if the procedure had been manual. Therefore, an R-language script was developed to automate the file linking process. Metadata containing date (day, month and year) and web browsing mode have been added. In the files generated by the web scraping process, missing values were coded with special characters, e.g. "-". All variables that are numeric in nature, such as price or the number of stars, had to be cleared of such special characters and replaced by the NaN value (not a number), and then converted into numeric values.

4. Data analysis

Data analysis is the process of checking the correctness of the obtained data as well as organizing and interpreting data and drawing conclusions based on them (test hypotheses or refute theories). The purpose of analyses is to generate knowledge about a given phenomenon.

4.1 Strategy for dealing with outliers

One of the first elements that was checked is the analysis of the data distribution. In order for the conclusions to be correct, the erroneous values resulting from the data acquisition process should first be removed. Most often this applies to measurement errors and errors when entering data (e.g. data entered by portal users). The erroneous values are out of reasonable scope of the possible values. Removing erroneous values in this way is a validation step. Outliers that are not the result of error but the immanent part of the phenomenon are not rejected. The threshold between reasonable values and outliers can be derived by a statistical method. The method adapted to that problem is presented below.

Such values will of course have an impact on the classic measures of the central tendency like the arithmetic mean or dispersion measures such as standard deviation. However, part of the data analysis is to draw conclusions as to what statistical measures can be used for the acquired data set.

There are many methods to search for outliers, but it is on the part of the researcher to decide what the value will be. Criteria for rejecting outliers can be broadly divided into three groups: constructing intervals on the basis of classical measures, constructing intervals on the basis of positional measures and using statistical tests.

The first one is the 3 sigm rule, which is based on the properties of the normal distribution. Under this rule, outliers are considered to be further away from the mean than the value of the three standard deviations. The second group includes a compartment based on data distribution quartiles:

$$[Q1 - k(Q3 - Q1), Q3 + k(Q3 - Q1)],$$

where k> 0 is determined by the researcher. If we denote IQR = Q3-Q1, then for k = 1.5 we have the following estimate.

Observations that may be suspected to be outliers are those that go beyond the range (Q1-1.5 IQR, Q3 + 1.5 IQR). Extreme outliers are those that go beyond the range (Q1-3 IQR, Q3 + 3 IQR). The third group of criteria includes, among others, Frank E. Grubbs test [Grubbs 1950].

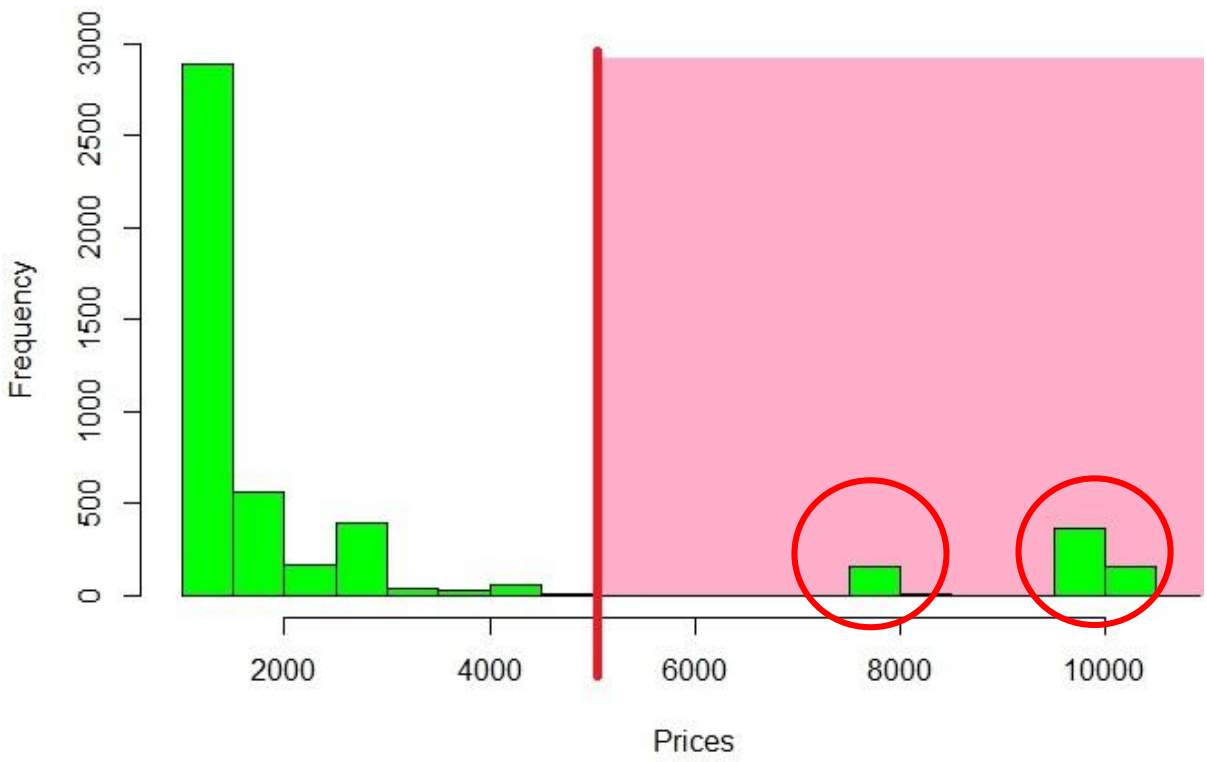
It turns out that in the case of data that are not characterized by normal distribution but have a large asymmetry and dispersion, the application of these rules often causes that values reliable for a given phenomenon are rejected or, on the contrary, unreliable values are accepted.

In the case of data obtained by the web scraping method, it can be noticed that the data distribution is characterized by right-sided asymmetry and strong dispersion. Details can be found in the use case 2.

It may turn out that classical methods of outlier detection are too general and therefore invalid. Thus, it may be recommended to use the histogram as a tool for detecting outliers. For instance, consider scrapped data on prices of tourist accommodation establishments. The maximum price amounted to 1 million PLN, i.e. three thousand times higher than the average price. Definitely, such an offer has an invalid price. It also indicates that other outliers may occur.

The figure below presents the proposed rule: find a value that divides the compact part of the histogram from the uncompact part. All values greater than the threshold value should be rejected from analysis or validated. Outliers are identified on the histogram and marked in red circles.

Figure 2. Recognizing outliers with histogram (price per person in PLN)



Source: Own work.

Therefore, we adopt the following rule for outliers:

If the histogram series with non-zero values are followed by empty series, then all subsequent non-zero series are treated as outliers.

On cleared data, basic measures of central tendency, dispersion, skewness and concentration are calculated in the classical and positional version as well as distribution analysis is performed. In the next step descriptive statistics should be made in various cross-sections, e.g. temporal, spatial and other, which result from the attributes of the data set. This allows to capture the specificity of the phenomenon and is necessary to generate statistics that will be the key to connecting the analysed

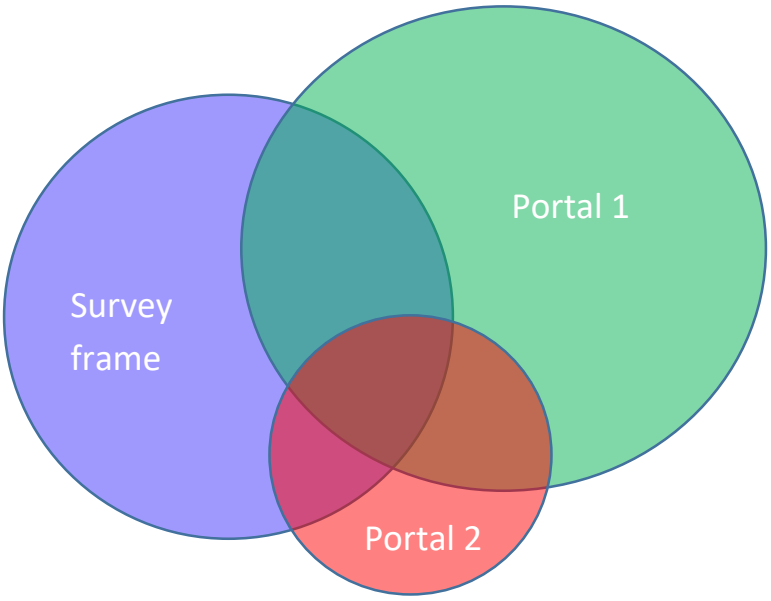
set with other sets. Such combining of sets often does not occur at the level of micro-data, and usually it is the level of micro aggregates, e.g. in a spatial or temporal cross-section.

4.2 Combining data sets

If the researcher wants to combine two sources of different frequencies (e.g. monthly and daily frequency), aggregation of high frequency data to the lower frequency level must be performed. As a result, two data sets of the same frequency will be obtained. The same situation occurs in different spatial aggregation levels of data sets. It is not direct linkage but it enables to apply flash estimate methods and spatial-temporal disaggregation methods.

If the researcher wants to link two sources of the same spatial-temporal granularity double counting problem will occur as presented below.

Figure 3. Overlapping of entities in survey frame and portals.



Source: Own work.

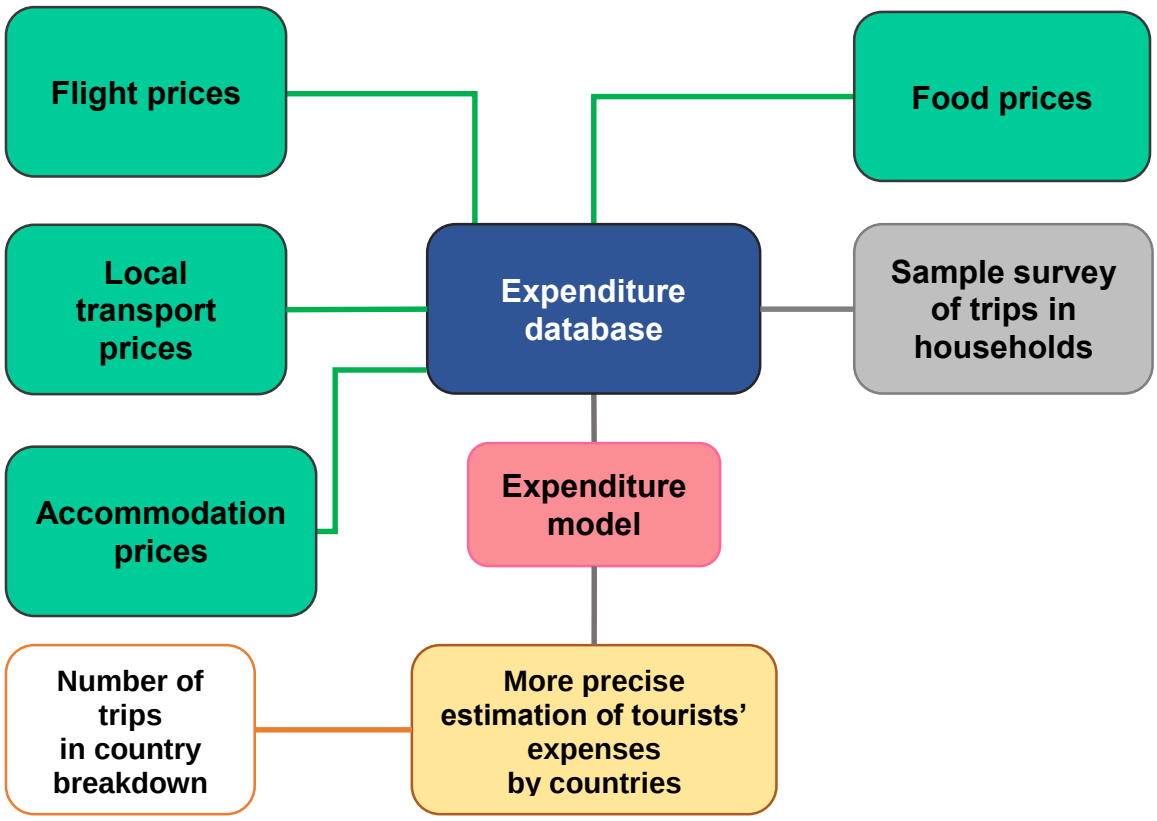
In such a case, to avoid double counting it is required to merge data sets by key variables such as address including postal code, street, etc. and name of entity. Regular expressions are a crucial tool to clean key variables and to make linkage accurate. The results of survey frame and scraped data linkage will be delivered in the next report.

4.3 Next steps – short run perspective

As part of the pilot project, ESSnet Big Data I developed a methodology for estimating travel in air traffic that was based on administrative data and web scraping. Naturally, the next step was to estimate travel costs based on data from web scraping. Therefore, it is necessary to collect information relating to various types of travel-related costs such as air tickets, food, accommodation, local transport, etc.

The diagram below presents the idea of combining data on travel costs from web scraping and from a sample survey as well as results on the number of trips by countries obtained by the method developed under ESSnet Big Data I.

Figure 4. Data combining in air traffic model



Source: Own work.

At present, data on accommodation costs are collected. Collecting of data on air ticket prices is in progress. After accumulating the appropriate amount of data, they will be analysed and their relationship with travel costs declared in the sample survey will be examined. As a result, it will allow to estimate the travel costs for countries that are poorly represented in the sample survey. Methodological details will be presented in subsequent reports.

5. Conclusions

Dynamic socio-economic changes, unexpected development of new technologies (in particular development of mobile technologies and user generated content) and cultural changes (such as globalization and expansion of the Schengen zone) in recent years often require identification of new tourism phenomena and tracking changes in tourism and tourists trends in a timely manner. Tourism statistics measuring supply and demand side of tourism is increasingly using innovative data sources that open up completely new opportunities to improve and enrich the existing system.

The issues presented in this document constitute a preliminary stage of work on the inclusion of big data in official statistics. On the basis of experience, literature overview and recommendations from ESSnet Big Data I, a universal web scraping method for extracting data from tourism-related websites has been developed.

As a result of the work carried out, portals with a high potential to enrich the results of traditional statistics characterized by a large amount of indexed data, high detail of data, frequent updating and increased users activity were identified. In next step those websites were grouped by two categories: global and local ones as well as according to information possible to obtain.

Due to differences in the legal aspects of web scraping in different countries, the legality of web scraping had been examined before the process started. Web scraping of websites should be non-invasive and used on the basis of good practices while maintaining statistical confidentiality and comply with data protection and copyright laws.

In the course of the work, an overview of the most popular methods of data extraction was carried out, as well as an analysis of their strengths and weaknesses. As a result of the analysis, an innovative approach to the web scraping method was presented, based on the concept of simulated browsing of tourist websites imitating the user's behaviour. One of the most important features of the presented approach is to ensure that the service is not able to distinguish whether the information on the website is viewed by a human being or it is an automatic machine. This is of great importance if we plan to collect data from a selected source in the long run.

Analysis of web scraping data requires the prior linking of multiple files, adding metadata, cleaning and formatting variables for the needs of the selected development environment used for statistical analysis. Data cleaning tools and statistical analysis (including strategy for dealing with outliers) implemented by WPJ allowed to minimize the occurrence of errors and their impact on quality of data.

The R environment was used in the project. During the analysis it turned out that the data as a result of web scraping for hotels.com and the selected local portal have right-sided asymmetric and bimodal distribution. Therefore, a unique method of rejecting outliers was developed. A number of other statistical analyses were carried out. It is a prerequisite for the use of data linking methods (bridge equations, time series model, etc.).

The development of a universal web scraping tool and methods of data analysis and combining are an iterative process and it is likely that in the course of further work they will require adjustments in terms of methods and variables. The final conclusions will be presented in the final report on the implementation of the project. We hope that this document will provoke a discussion about web scraping, in particular its role and possibilities of using it in tourism statistics, which can also benefit from many other sources.

Appendixes

Use case 1

The booking.com service requires the user to perform a number of actions in order to present the result of the booking.com service in the browser window. Analysis of the website in terms of its internal structure indicates that the website operates in an asynchronous, dynamic way and requires user interaction to work properly. Additionally, working data are stored in the form of session variables, which must be available all the time while working with the service. Selected parameters are transferred to the service in the form of coded values of URL variables.

In connection with the above, algorithm steps which allow to store data in a form useful for further processing have been developed.

Algorithm:

1. From the configuration file the following data are loaded: city, dates from/to check-in, number of rooms, number of adults and number of children, type of accommodation.
2. On the basis of loaded data, in an interactive way, the program fills in the appropriate fields of the form and sends the form to the server.
3. After refreshing the page, new items for selection appear. In this step we choose filters in terms of accommodation type - the selection of the list of accommodation establishments for filtering comes from the configuration file, where appropriate establishments are selected interactively.
4. After reloading the page, the page view is moved downwards, and the number of pages to be processed is detected.
5. The current page and all parameters that have been loaded from the configuration file are saved in a text file (with the number of the current subpage).
6. The next page in the paging menu is selected interactively; after refreshing we go back to the point 5 provided that this is not the last page.
7. When the last page is loaded, the resources are released and the program is terminated.

Additional application functionalities not included in the algorithm:

1. The application saves working data for each downloaded page on a given day and all subpages; if for some reason it is not possible to complete the download of all data, the application with a delay of a few seconds, repeats its attempt to download the data that could not be downloaded.
2. The application saves a full copy of the HTML source data together with the data; this makes it possible to process the archived data from a given period of time.
3. Due to the dynamics of the service and asynchronous data injection, there was a need to develop a solution that will ensure that the data will be saved and processed only when all data are loaded on the website.

Use case 2

Web scraping of Hotels.com has been conducted in a testing mode since November 2018 for the capital region of Poland. Since mid-February 2019 it has been performed on a regular basis for all of the regions (NUTS 2 level). It provides information about:

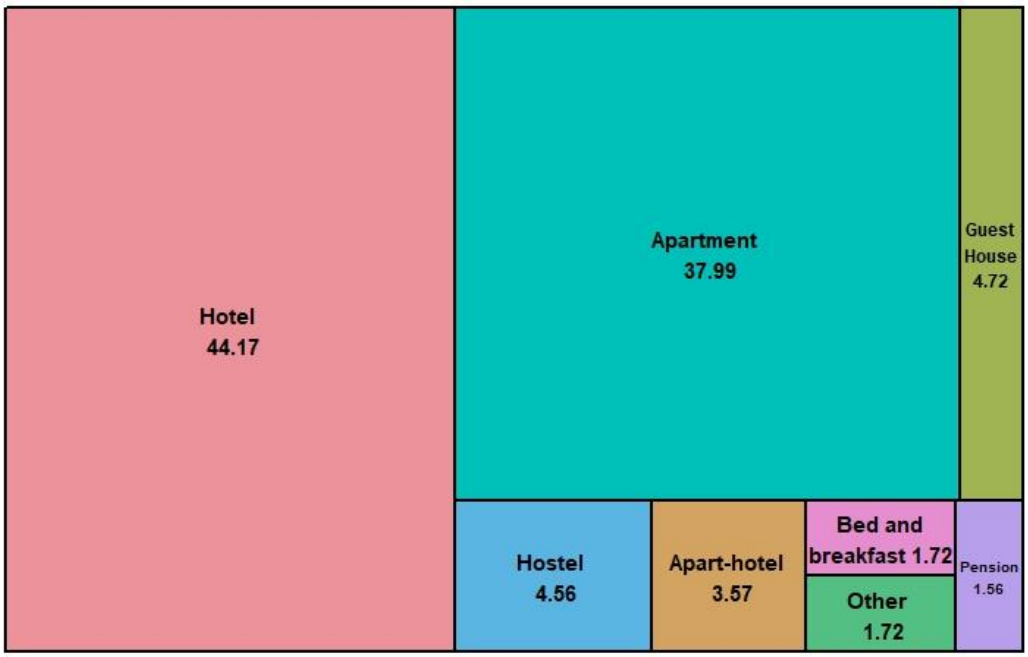
- offer Id,
- hotel Id,
- hotel name,
- type of accommodation,
- address,
- price (per night per person),
- star rating,
- guest reviews,
- number of guest reviews.

Files were merged into one database and metadata on the day, month, year and type of web scraping was added. Data were cleaned and formatted to allow analysis.

All of the figures pertain to capital region of Poland and are based on scrapped data solely. At this moment, there are no price-based official statistics pertaining to tourist accommodation establishment to compare it with scrapped data.

As shown in Figure 5, hotels provide almost half of the offers (44.17%), followed by apartments (37.99%). Other groups do not exceed 5% of the total number of offers.

Figure 5. Tree map of overnight offers from web scraping by accommodation type



Source: Own work.

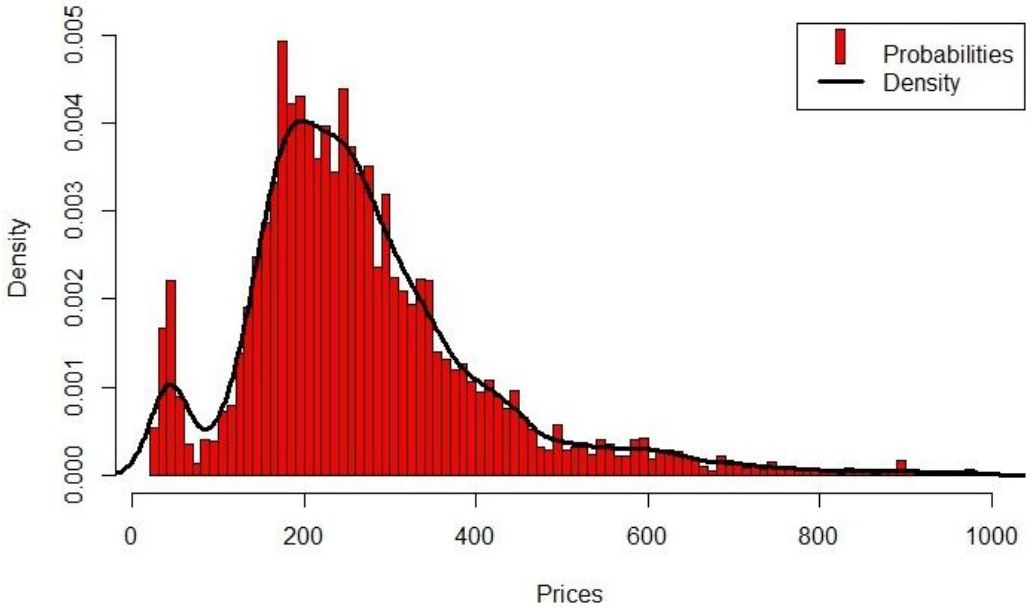
A preliminary analysis of prices and the number of offers was carried out with respect to accommodation type and destination.

First, it was examined whether there are outliers, which arise as a result of errors in data acquisition and processing, significantly hindering and in the extreme case preventing correct analysis.

Prices were characterized by highly dispersed distribution. Preliminary rejection rules based on three standard deviations and three quartile ranges were applied. In the first case, many unreliable values were not discarded according to the rule, while in the second case, the opposite was true. That is why a unique approach was used, which in subsequent periods will be checked as to its legitimacy. It was assumed that outliers occurred when the maximum price exceeded the 99th percentile twice. Based on the histogram of prices a threshold dividing "solid" part of the histogram from "lonely islands" was chosen. All values above the 99th percentile were removed and not used for further analysis.

It turned out that the distribution of prices of tourist accommodation establishments is right-skewed and surprisingly bimodal (based on density estimation). Skewness coefficient amounted to 13.85, which is twice as much as in the case of log-normal distribution, which indicates really high asymmetry.

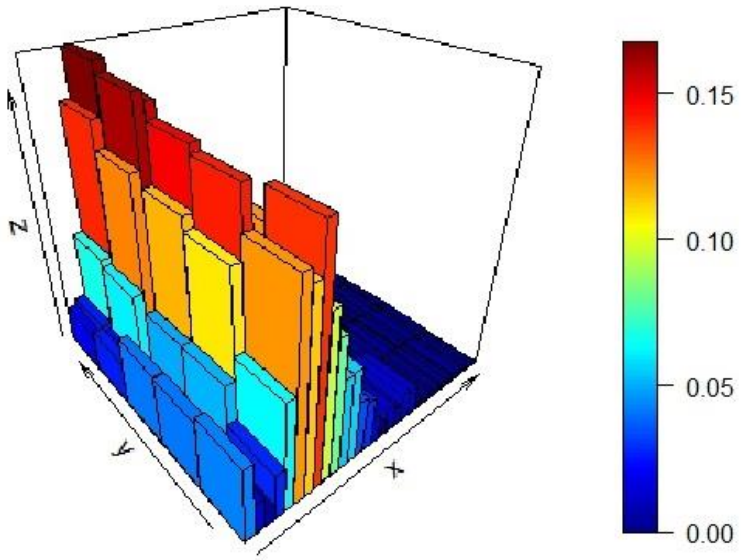
Figure 6. Histogram of prices with density estimation



Source: Own work.

In the next figure, x axis pertains to price ranges, y axis refers to consecutive months (from November 2018 to March 2019), and there are relative frequencies on z axis.

Figure 7. Histogram of prices by months (November 2018 to March 2019)

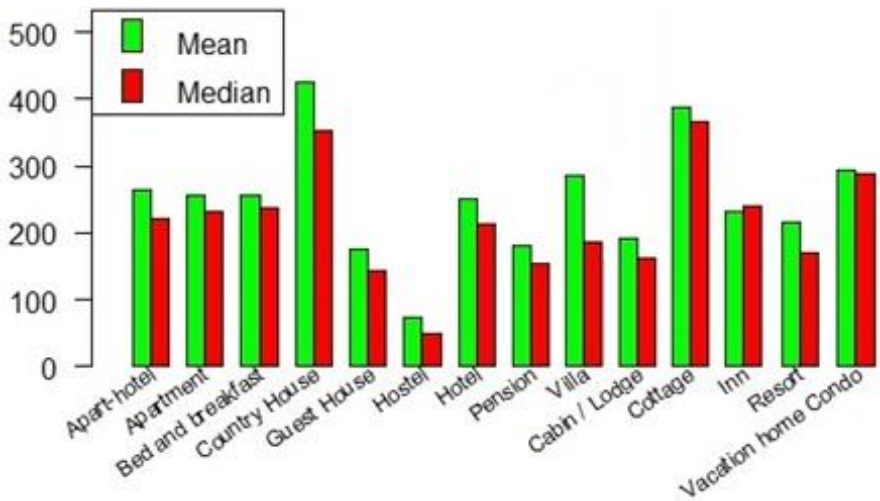


Source: Own work.

It can be observed that the distribution of prices of tourist accommodation establishments is right-skewed and bimodal in each of the analysed months.

To solve the puzzle – why bimodality is observed – basics statistics (mean and median price) were calculated with respect to accommodation type and region. In such a breakdown, significant differences in means or median may indicate possible solution.

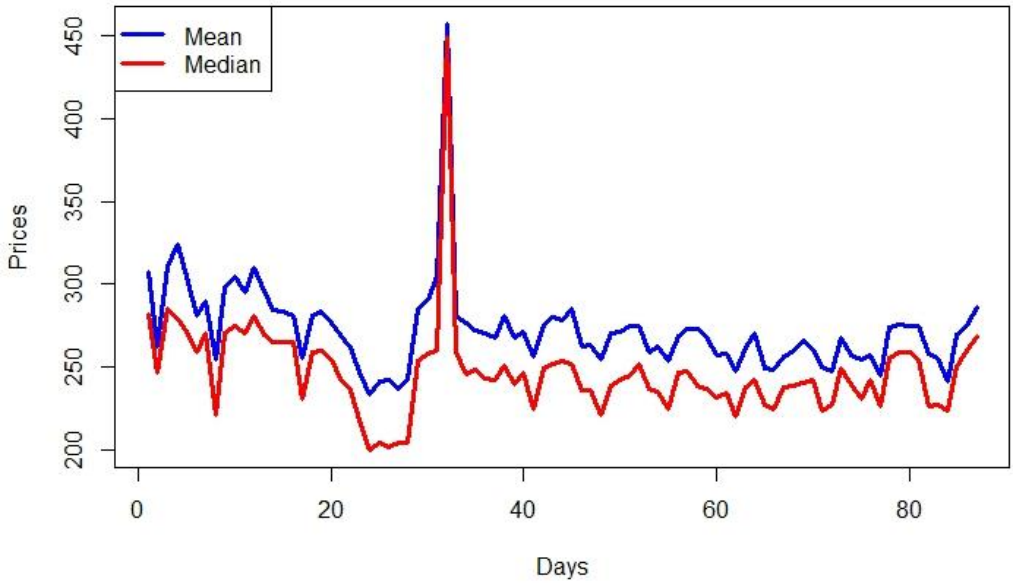
Figure 8. Median and mean prices by accommodation type



Source: Own work.

It was discovered that hostels, which represent 4.5% of the offers, offer significantly lower prices than tourist accommodation establishments on average. It turned out that hostels offers are the substantial part of offers among offers with prices up to 1st percentile of prices. This is the reason why such a high density of offers can be observed at a very low level of prices. Median and mean prices trends were examined. Although these central tendency measures are usually used in the case of asymmetric and symmetric distribution respectively, it turns out that their trends are very similar. In the next figure, x axis pertains to consecutive days (from the end of November 2018 to the beginning of March 2019), y axis refers to prices.

Figure 9. Median and mean prices trends



Source: Own work.

In fact, the correlation coefficient amounted to 0.988. This is a statistically significant result at the level of $p < 0.001$. It is New Year's Eve. It turned out that prices shifted a lot due to increased demand just before New Year's Eve, while the number of offers decreased. This is the reason for such a peak in the diagram.

Data sets linkage is on preliminary stage. After cleaning a postal code in the survey frame and in scrapped data and after counting establishments on postal code level it has been discovered that number of establishments not covered by the survey frame amounted to 10,7%.

Use case 3

Web scraping of flight ticket on Booking.com has just started. The search engine for flight tickets on Booking.com is provided by kayak.com. The web scraping was set with parameters:

- one adult person in the economy class
- one-week trip
- flying from Warsaw to the destinations:
    - Mexico City Airport (MEX), Mexico City, Mexico,
    - Buenos Aires International Airport (EZE), Buenos Aires, Argentina,
    - La Paz Airport (LPB), La Paz, Bolivia,
    - Brasilia Airport (BSB), Brasília, Brazil,
    - Santiago Airport (SCL), Santiago, Chile,
    - Quito Airport (UIO), Quito, Ecuador,
    - Georgetown International Airport (GEO), Georgetown, Guyana,
    - Bogota Airport (BOG), Bogota, Colombia,
    - Asuncion Airport (ASU), Asuncion, Paraguay,
    - Lima International Airport (LIM), Lima, Peru,
    - Paramaribo Airport (PBM), Paramaribo, Suriname,
    - Montevideo Airport (MVD), Montevideo, Uruguay,
    - Caracas Airport (CCS), Caracas, Venezuela,
    - Tokyo Haneda Airport (HND), Tokyo, Japan,
    - Shanghai Pudong International Airport (PVG), Shanghai, China.

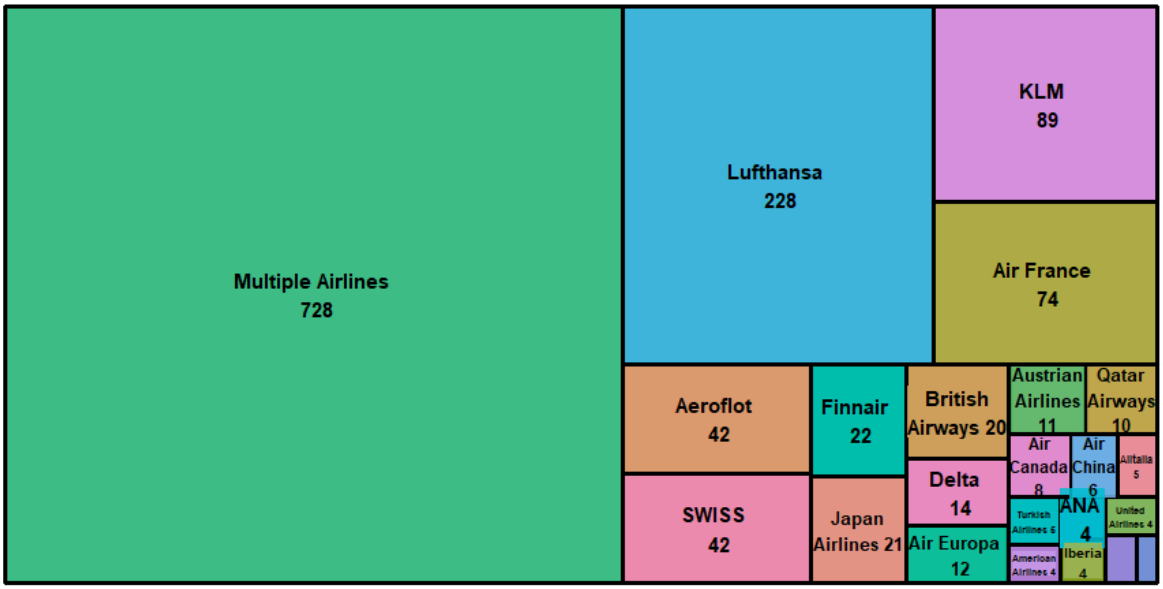
Query results provided information about:

- airlines,
- stops (hub airports),
- departure time,
- arrival time,
- flight duration,
- price in USD,
- deal details (e.g. “Only 1 seat left at this price”).

The search engines results turned out to be more complex than for accommodation offers. Web scraping output files varied with the number of columns because some additional information was scrapped (e.g. deal details), which increased the bulk of data for a single offer. Files were merged into one database which required a bit of engineering due to different number of columns in each file. Data were cleaned and formatted to allow analysis. Database contains 1358 offers generally representing South America. Since there are 15 destinations it is rather small amount of data. Thus, only cautious conclusions might be drawn.

The next figure presents the number of offers with respect to airlines. Some offers contain flights serviced by different airlines and those are labelled with “Multiple Airlines”.

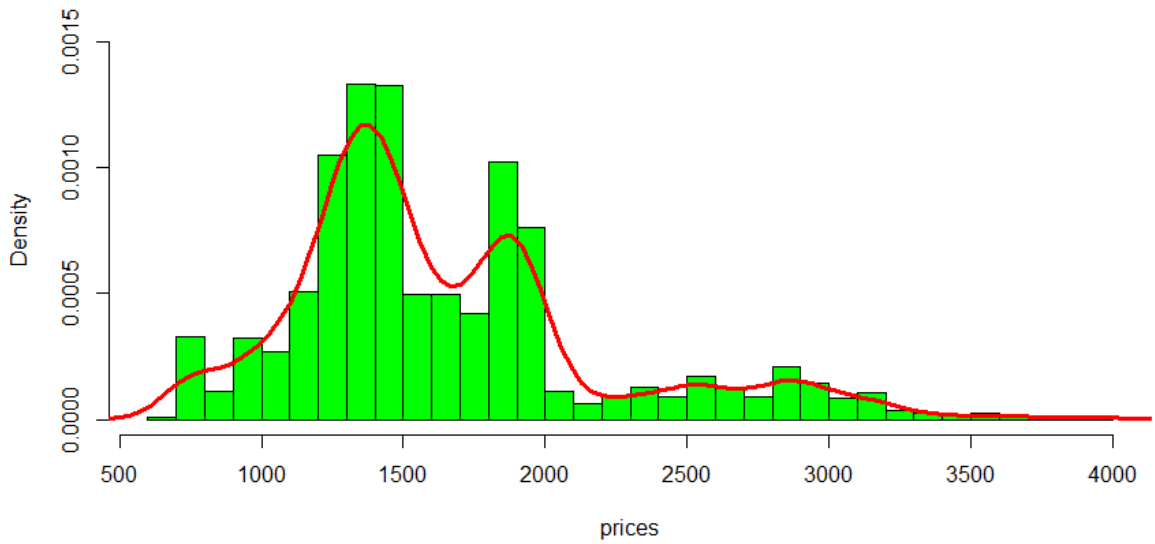
Figure 10. Tree map of flight ticket offers from web scraping by airline



54% of offers were covered by multiple airlines, 17% solely by Lufthansa, 7% by KLM and 5% by Air France.

Some offers were discarded from price analysis since they were found to be the outliers. The histogram of prices (figure below) shows that the distribution of prices is right-skewed.

Figure 11. Histogram of prices with density estimation



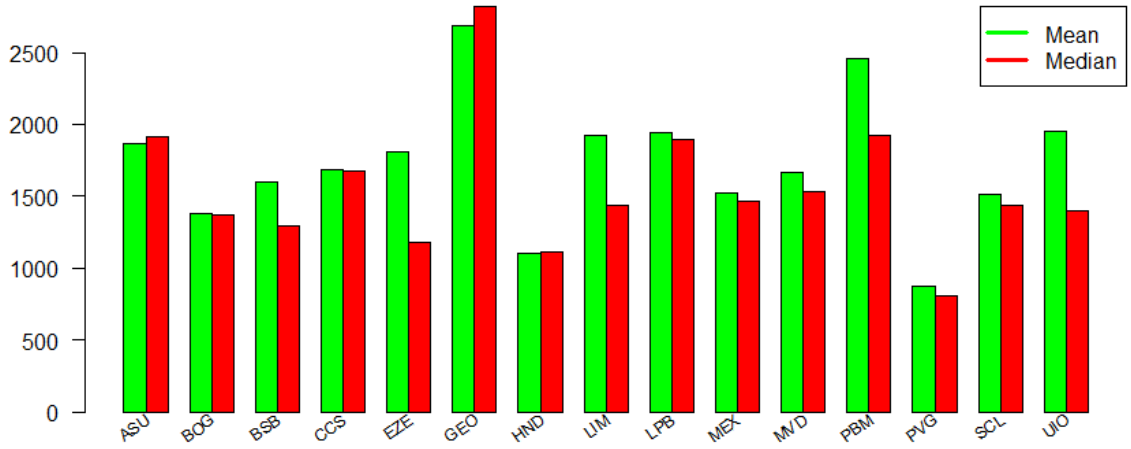
Source: Own work.

The second peak in the histogram around price of 1900 USD is formed by Georgetown and Paramaribo. These destinations require usually three stops while for the rest of airports it is usually up to two stops.

Although they are placed geographically closer to Warsaw, the distance for the aircraft is longer than for, e.g. Buenos Aires. Thus, the number of stop may be an explanatory variable for the costs of tickets.

Mean and median prices were compared with respect to the destination.

Figure 12. Median and mean prices by destination airport



Source: Own work.

In general the median price seems to be lower than the mean price. The Pearson correlation coefficient between median and mean price amounted to 0.85. It is not as strong as for accommodation prices.

More detailed analysis will be carried out after gathering relevant bulk of data.

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Glossary

CSS (Cascading Style Sheets) is a style sheet language used for describing the presentation of a document written in a markup language. CSS is a cornerstone technology of the World Wide Web.

CSV (comma-separated value format) is a delimited text file that uses a comma to separate values. A CSV file stores tabular data (numbers and text) in plain text. Each line of the file is a data record. Each record consists of one or more fields, separated by commas.

DOM (Document Object Model) is a way of representing complex XML and HTML documents in the form of an object model. This model is independent of the platform and programming language.

GUI (Graphical User Interface) is a method of the user's communication with a computer using graphic elements.

XML (Extensible Markup Language) is a universal markup language designed to represent different data in a structured way. It is platform-independent, which facilitates the exchange of documents between different systems.

HTML (HyperText Markup Language) is a hypertext markup language that allows one to describe the structure of information contained within a web page, giving meaning to individual fragments of text – forming hyperlinks, paragraphs, headers and lists - and embedding file objects in the text

HTTP (The Hypertext Transfer Protocol) is an application protocol for distributed, collaborative, hypermedia information systems. HTTP is the foundation of data communication for the World Wide Web, where hypertext documents include hyperlinks to other resources that the user can easily access.

JSON (JavaScript Object Notation) is a text format for saving and exchanging data.

SSL (Secure Socket Layer) is cryptographic protocols designed to provide communications security over a computer network. Websites can use SSL to secure all communications between their servers and web browsers.

XLSX (Office Open XML) is a zipped, XML-based file format developed by Microsoft for representing spreadsheets, charts, presentations and word processing documents.

Web scraping (also known as Screen Scraping, Web Data Extraction, Web Harvesting, etc.) is a technique used to extract large amounts of data from websites.